

Project_Information

GND — GND

Title **BK6970**			
Size: Tabloid	Number: D2088000	Revision:	
Date: 21/10/2026	Time: 11:44:40	Sheet 1 of 4	
Drawn by: ***Blaz Kvas***			

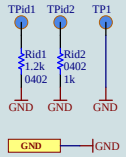
Luxonis Holding
1925 Harmony Park Drive
Westminster, CO
80234
United States

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Project: ***BK6970***
Current Revision:

BK6970 *Revision History:*

Date	Revision		Reason for Change	Changes Implemented
06/03/2026	R0	D0		
24/07/2026	R1	D0	- Make it look nice and pretty.	- Change the color of PCB to purple.

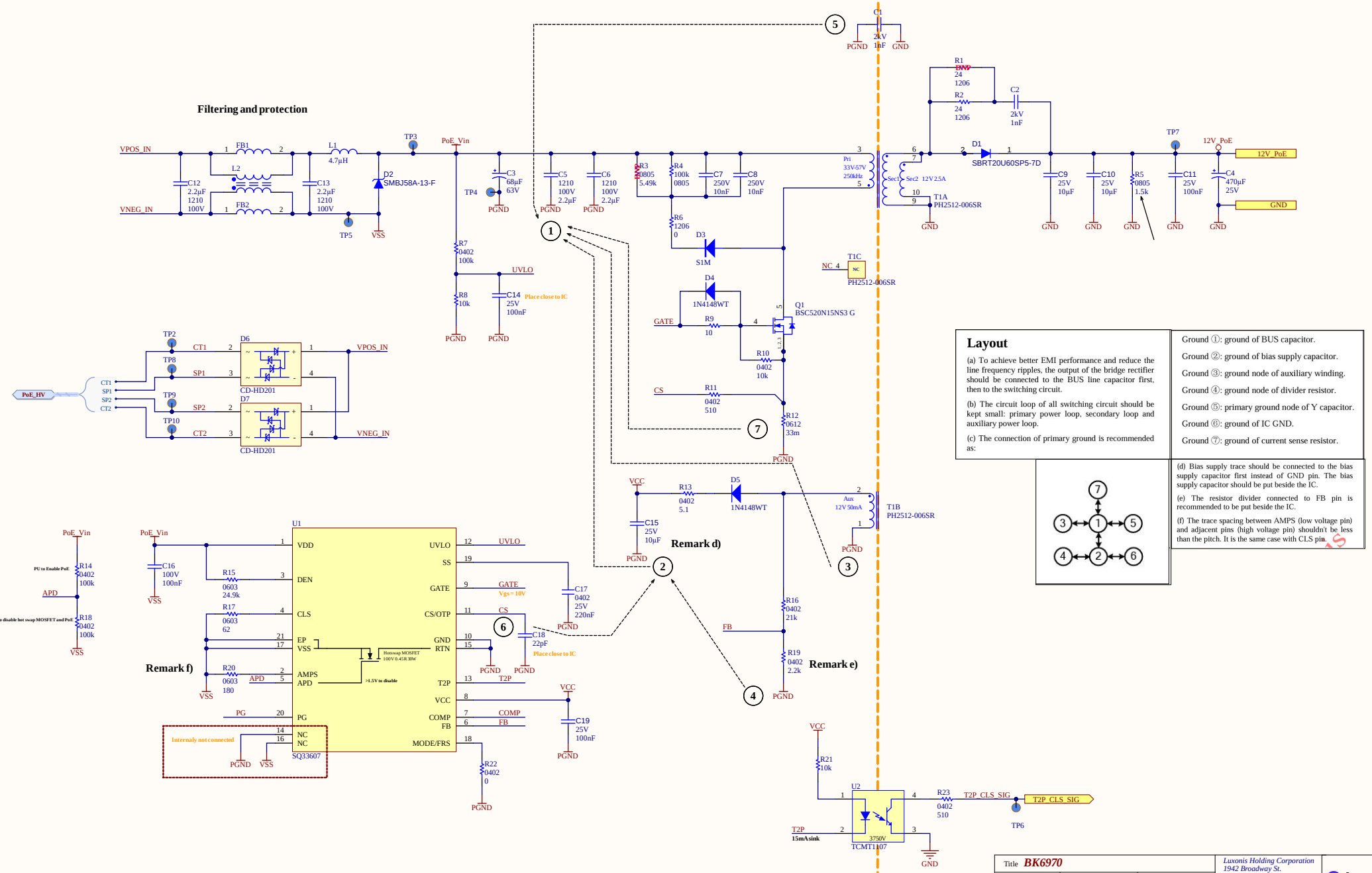


Rid1	Rid2
1K0	DNP
1K2	DNP
1K5	DNP
1K8	DNP
2k2	DNP
2k7	DNP
3k3	DNP
3k9	DNP
4k7	DNP
5k6	DNP
6k8	DNP
8k2	DNP
10k	DNP
12k	DNP
15k	DNP
18k	DNP
22k	DNP
27k	DNP
33k	DNP
39k	DNP
47k	DNP
56k	DNP
68k	DNP
82k	DNP
100k	DNP
120k	DNP
150k	DNP
180k	DNP
220k	DNP
270k	DNP
330k	DNP
390k	DNP
470k	DNP
560k	DNP
680k	DNP
820k	DNP
1k0	1k0
1k2	1k0
1k5	1k0
etc.	etc.

Rid1 / Rid2	RxDxVx
1k0 / DNP	R0D0V0
1k2 / DNP	R1D0V0

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Layout

(a) To achieve better EMI performance and reduce the line frequency ripples, the output of the bridge rectifier should be connected to the BUS line capacitor first, then to the switching circuit.

(b) The circuit loop of all switching circuit should be kept small: primary power loop, secondary loop and auxiliary power loop.

(c) The connection of primary ground is recommended as:

Ground ①: ground of BUS capacitor.

Ground ②: ground of bias supply capacitor.

Ground ③: ground node of auxiliary winding.

Ground ④: ground node of divider resistor.

Ground ⑤: primary ground node of Y capacitor.

Ground ⑥: ground of IC GND.

Ground ⑦: ground of current sense resistor.

(d) Bias supply trace should be connected to the bias supply capacitor first instead of GND pin. The bias supply capacitor should be put beside the IC.

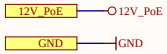
(e) The resistor divider connected to FB pin is recommended to be put beside the IC.

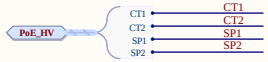
(f) The trace spacing between AMPS (low voltage pin) and adjacent pins (high voltage pin) shouldn't be less than the pitch. It is the same case with CLS pin.

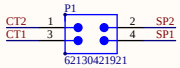
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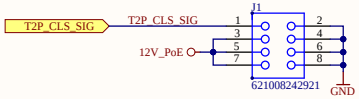
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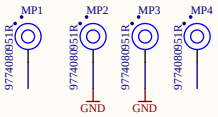
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